

Jenkins



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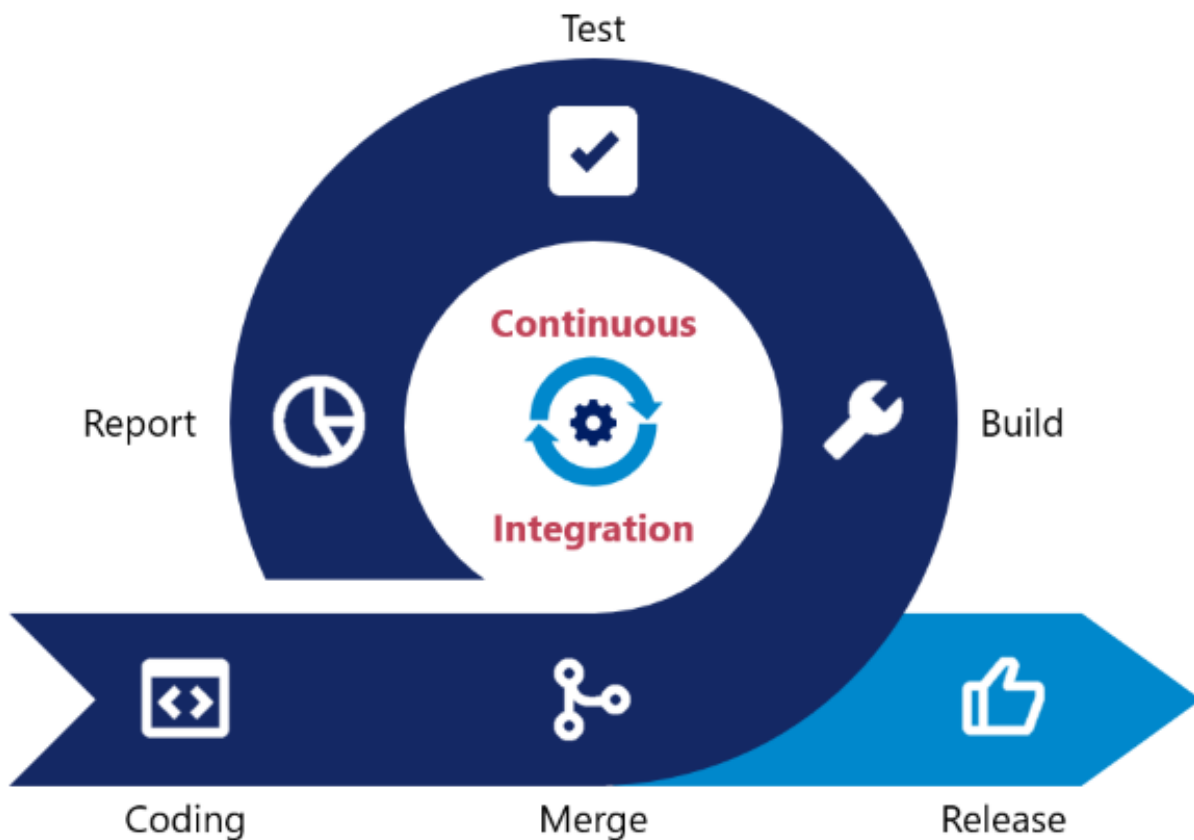
Jenkins

هي CI/CD Tool ، طيب هو أي ال CI/CD دا ؟

CI اختصار ل continuous integration :

هي اني بخلي عملية ال release بتاع ال New Feature او أي Change يكون بشكل Automated .

الفكرة اني ببقا عندي Central Repo عليه ال Code واي change تحصل علي ال repo دا بعد كدا ال CI/CD من خلالها لما يحصل Change علي ال Repo دا يحصل Build ال Build داه يعمل 3 حاجات : هيعمل Test و artifacts و push لل artifacts دا

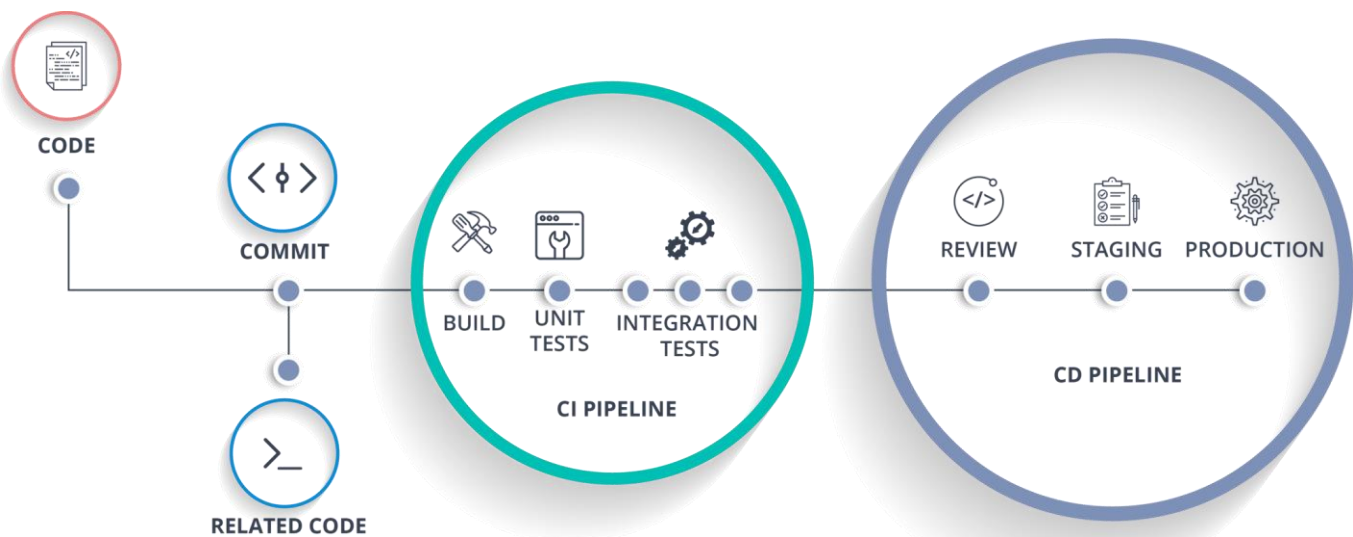


CD اختصار ل continuous delivery/deployment

ال continuous deployment : لم ال Build يحصل ال deploy بيكون بشكل Automated
علي كل حابه عندي سواء ال Testing او ال Production

لكن ال

ال continuous delivery : بيحي عند ال Production ويبستني Approve من شخص عشان ال
push او ال deploy يحصل علي ال Production



ال Tools الخاصة بال CI/CD كثير جدا وعشان كذا مهم انك تكون فاهم ال Concept كويس

Open Source



SaaS



Cloud Services



طيب ليه نستخدم ال Jenkins ؟

1- انها open Source

2- انها Java Based ودا بيساعد في انها تشتغل علي أي OS

3- فيه منها ال Free وال Enterprises

4- ليها اكثر من 1500 Plugins

5- ليها Community كبير

6- SCM Integration

: Install Jenkins

عشان اعمل Install لل Jenkins في اكثر من طريقه

Windows-1

Linux-2

Docker-3

K8s-4

ومن خلال اللينك دا تقدر تشوف الطريقه المناسبه

<https://www.jenkins.io/doc/book/installing>

بعد م بتعمل Install هنبدأ نعمل الخطوات دي عشان نبدا نشتغل علي ال Jenkins

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

بعمل Unlock لل Jenkins من خلال ال admin password ودا تقدر تعرفه من اكثر من مكان وهو بيكون كاتبلك ال Path

```
[root@192 secrets]# cat /var/lib/docker/volumes/jenkins_home/_data/secrets/initialAdminPassword
546aff02a41d47e484298e8758e1ea93
[root@192 secrets]#
```

```
[root@192 ~]# sudo docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
546aff02a41d47e484298e8758e1ea93
[root@192 ~]#
```

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

.....

Continue



Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

بعد کدا تقدر تعمل Custmize ب انك تختار Plugins معينه او install کل ال Plugins

Getting Started

✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding	on Framework
✓ Timestampers	Workspace Cleanup	Ant	Gradle	(JAF) API ** JAXB ** SnakeYAML API ** JSON Api ** Jackson 2 API ** commons-text API ** Pipeline: Supporting APIs ** Plugin Utilities API ** Font Awesome API ** Bootstrap 5 API ** JQuery3 API ** ECharts API ** Display URL API ** Checks API ** JUnit ** Matrix Project ** Resource Disposer ** - required dependency
Pipeline	GitHub Branch Source	Pipeline: GitHub Groovy Libraries	Pipeline Graph View	
Git	SSH Build Agents	Matrix Authorization Strategy	PAM Authentication	
LDAP	Email Extension	Mailer	Dark Theme	

Jenkins 2.452.2

بييدا يعمل Install لل Plugins

Create First Admin User

Username

admin

Password

•••••

Confirm password

•••••

Jenkins 2.452.2

[Skip and continue as admin](#)[Save and Continue](#)

بيطلب منك تعمل admin user وممكن تعمل Skip

Instance Configuration

Jenkins URL:

http://192.168.93.130:8080/

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.452.2

Not now

Save and Finish

بعد كذا بيديك ال URL

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

Start using Jenkins

وبيقولك ان ال Setup تمام وتقدر تبدأ شغل علي ال Jenkins

System

Home directory ?

By default, Jenkins stores all of its data in this directory on the file system

/var/jenkins_home

هنا دا ال Path ال Jenkins بيخزن فيه كل حاجه تخصه

Plain text [Preview](#)

of executors

2

Label

دا يعني اقدر اعمل كام Run لكam job في نفس الوقت (يعني عدد ال pipeline ال اقدر اعملها run في نفس الوقت).

Labels

دي لو علوز ادي ال Jenkins اسم معين

Usage ?

Use this node as much as possible

Use this node as much as possible

Only build jobs with label expressions matching this node

5

: Use this node as much as possible

دي لو محدش محدد ال pipeline يعمل run فين يجي يعمل run علي ال Master دي

: Only build jobs with label expressions matching this node

دي لا ميعملش Run غير لو انا محدده ان ال pipeline يحصله run في ال node دي

Build Queue



No builds in the queue.

Build Executor Status



1 Idle

2 Idle

Quiet period ?

5

هنا لو عندي pipeline بيدخل عطل على ال executor ولا يستني مثلا 5 ثواني اهي ال 5 ثواني دي هي ال Quiet period

SCM checkout retry count

2

لو الدنيا حصلها failed وهو بيعمل clone من ال github يحاول كام مره تاني

Pipeline Speed / Durability

Default Speed / Durability Level ?

None: use pipeline default (MAX_SURVIVABILITY)

None: use pipeline default (MAX_SURVIVABILITY)

Performance-optimized: much faster (requires clean shutdown to save running pipelines)

Less durability, a bit faster (specialty use only)

Maximum durability but slowest (previously the only option)

Usage Statistics

☒ Help make Jenkins better by sending anonymous usage statistics and crash reports to the Jenkins project ?

ال Max دي لو ال Jenkins وقع ورجع هيكمل من مكان ما يوقف

ال less دي لو وقع هيرجع بيبدأ من الأول

ال performance دي ما بين الاتنين

Timestamper

System clock time format ?

''HH:mm:ss''

Elapsed time format ?

''HH:mm:ss.S''

☐ Enabled for all Pipeline builds ?

دي الفورمات بتاع التاريخ

Dashboard > Manage Jenkins > Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Search available plugins

Install

Install	Name ↓	Released
☐	Command Agent Launcher 107.v773860566e2e Agent Management Allows agents to be launched using a specified command.	6 mo 2 days ago
☐	Oracle Java SE Development Kit Installer 73.vddf737284550 Allows the Oracle Java SE Development Kit (JDK) to be installed via download from Oracle's website.	6 mo 6 days ago
☐	JSch dependency 0.2.16-86.v42e010d9484b_ Library plugins (for use by other plugins) Miscellaneous Jenkins plugin that brings the JSch library as a plugin dependency, and provides an SSHAuthenticatorFactory for using JSch with the ssh-credentials plugin.	1 mo 3 days ago

دي ال plugins ال موجوده وال اقدر استخدمها في ال Jenkins

Security

Security

Secure Jenkins; define who is allowed to access/use the system.

Credentials

Configure credentials

Credential Providers

Configure the credential providers and types

Users

Create/delete/modify users that can log in to this Jenkins.

In-process Script Approval

Allows a Jenkins administrator to review proposed scripts (written e.g. in Groovy) which run inside the Jenkins process and so could bypass security restrictions. 1 scripts pending approval.

لو من ال Manage اختارت ال Security هيعمل list بكل ال اقدر اعمله ومنها :

--



Security

Authentication

☐ Disable "Keep me signed in" ?

Security Realm

Jenkins' own user database

Delegate to servlet container

Jenkins' own user database

LDAP

Unix user/group database

None

Save

Apply

في ال Security هلاقي ال Authentication وال Authorization

ال security Realm دي ال DB ال بيتخزن فيها ال Users

--

Dashboard > Manage Jenkins > Security

Jenkins' own user database

☐ Allow users to sign up ?

Authorization

Logged-in users can do anything

Anyone can do anything

Legacy mode

Logged-in users can do anything

Matrix-based security

Project-based Matrix Authorization Strategy

دي الحاجات ال بعمل Manage بيها لل Authorization بس فيه Plugin بتساعدني في اني اعمل Manage بطريقه افضل لل Authorization ال plugin دي اسمها

Role-Base Authorization Strategy

--

 **Jenkins**

Search (CTRL+K) ? Admin ▾

Dashboard > Manage Jenkins > Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

rol

Install

Install	Name ↓	Released
☒	Role-based Authorization Strategy 727.vd344b_eec783d Security Authentication and User Management Enables user authorization using a Role-Based strategy. Roles can be defined globally or for particular jobs or nodes selected by regular expressions.	1 mo 25 days ago

بدخل علي ال plugin وبعملها install

--

Jenkins' own user database

☐ Allow users to sign up ?

Authorization

Logged-in users can do anything

Anyone can do anything

Legacy mode

Logged-in users can do anything

Matrix-based security

Project-based Matrix Authorization Strategy

Role-Based Strategy

Markup Formatter

لما ارجع علي ال Authorization بلاقيها ظهرت معايا

--

Security

- | | |
|--|---|
|  Security
Secure Jenkins; define who is allowed to access/use the system. |  Credentials
Configure credentials |
|  Credential Providers
Configure the credential providers and types |  Manage and Assign Roles
Handle permissions by creating roles and assigning them to users/groups |
|  Users
Create/delete/modify users that can log in to this Jenkins. | |

ولو رجعت علي Security بلاقي tap جديده ظهرت خاصه بيها

--

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Manage Roles

Global roles

Role	Overall	Credentials			Agent					Job					Run													
	Administer	Read	Create	Delete	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move	Read	Workspace	Delete	Reply	Update	Configure	Create	
 admin	☒	☐	☐	☐	ManageDomains	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
 DevOps	☒	☐	☐	☐		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
 Testing	☐	☐	☐	☐		☐	☐	☐	☐	☐	☒	☐	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Save

Apply

لو دخلت عليها بلاقي ان فكرتها role

بتعمل Role معينه وبعد كذا تعمل

--

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Assign Roles

Global roles

User/Group	Devops	Testing	admin
Anonymous	☐	☐	☐
Authenticated Users	☐	☐	☐
Admin	☐	☐	☒

Add User

Add Group

User ل Assign

من Manage هندخل علي Credentials في ال Tap الخاصه بال Security ودي مكان بنخزن فيه ال user and password عشان ميظهرش في ال Pipeline

New credentials

Kind

Username with password

Username with password

GitHub App

SSH Username with private key

Secret file

Secret text

Certificate

Username ?

دي الحاجات ال ممكن استخدامها وليكن هعمل user and password

--

Username ?

☐ Treat username as secret ?

Password ?

ID ?

Description ?

Create

بعد م بتكتب ال user وال password فيه حاجه اسمها ID

ال ID هو ال انت بتتده عليه داخل ال pipeline عشان يستخدم ال Credentials دي

[+ New Item](#)[Build History](#)[Manage Jenkins](#)[My Views](#)[Open Blue Ocean](#)

Build Queue

No builds in the queue.

Build Executor Status

1 Idle

Credential Providers

Providers

All available

Types

All available

Restrictions

Add

Save

Apply

من Manage هندخل علي Credential Providers في ال Tap الخاصه بال Security
ال Credential Providers نفس فكره ال Credential لكن بيتم حفظها علي ال Cloud

Users

[+ Create User](#)

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID	Name
 mostafa	Mostafa Mahmoud 

من Manage هندخل علي Users في ال Tap الخاصه بال Security ومنها اقدر اضيف user

 **Jenkins**

Dashboard >

 New Item

 People

 Build History

 Manage Jenkins

 My Views

ال new item هبدا اعمل create لل pipeline

Enter an item name

» A job already exists with the name 'freestyle'

**Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

**Multi-configuration project**

ال freestyle نوع بسيط من ال pipeline

Plain text [Preview](#)

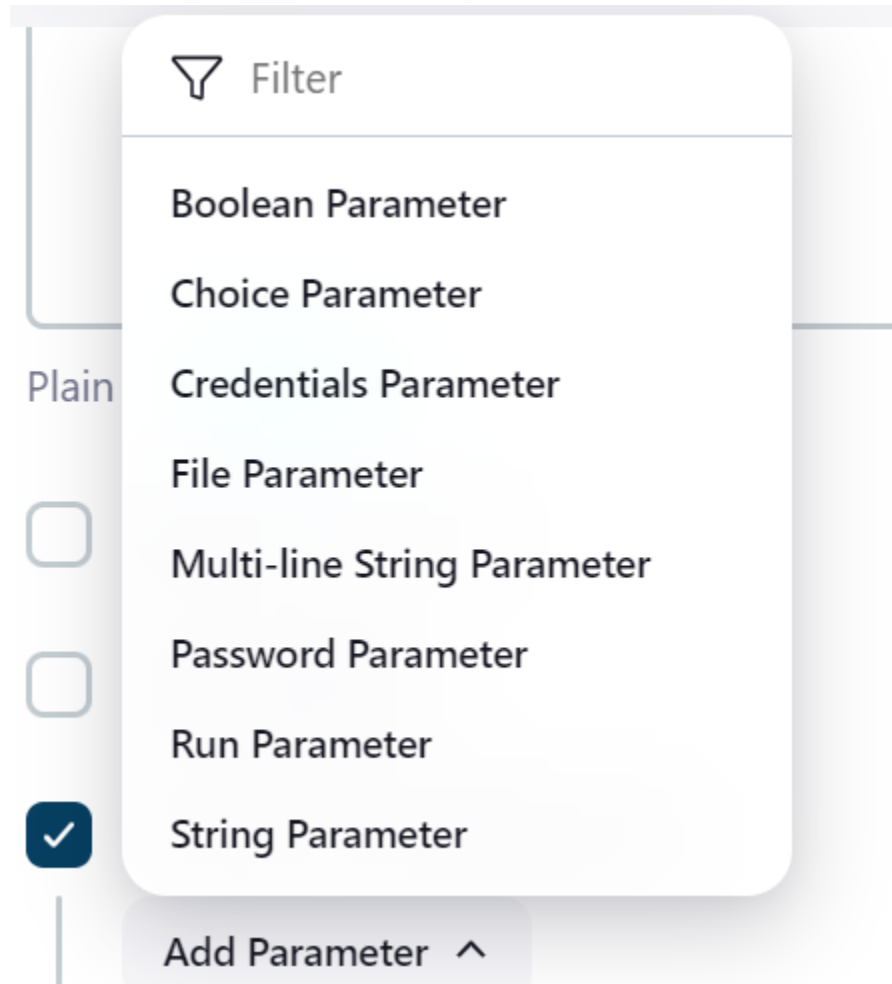
- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ This project is parameterized ?
- ☐ Throttle builds ?
- ☐ Execute concurrent builds if necessary ?

Advanced ▼

1-discard old builds : بتحدد عدد او أيام معينه يسبلك فيها ال old build ودي حاجه مفيدة لأنه لو فضل يسيب كل ال old build ع المدي البعيد هيعمل مشكله في ال storage مثلا هتتملي بسرعه لأنها فالآخر هي files بتترمي ع ال Jenkins

2-GitHub project : دي لو محتاج انك تعمل لينك لل project

parameterized-3 : دا مثلا قبل م تعمل run لل pipeline يكون فيه option بتختاره

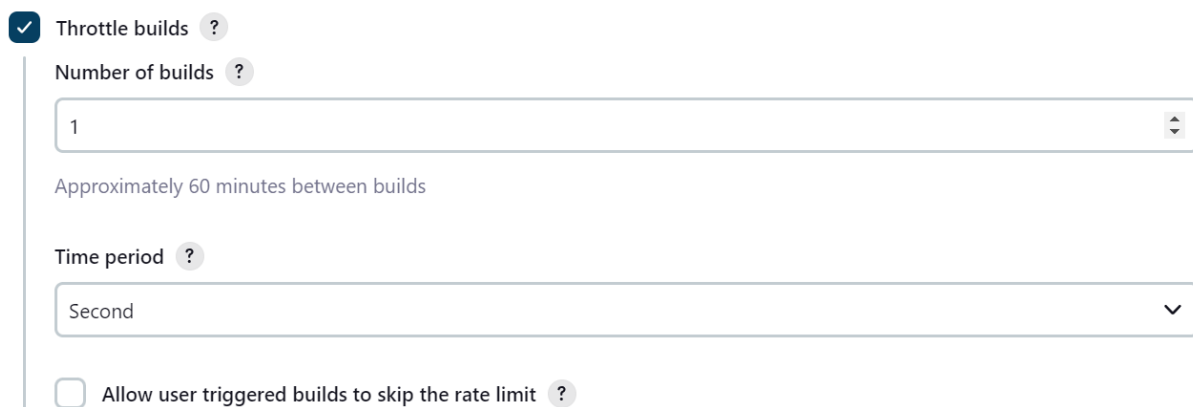


Filter

- Boolean Parameter
- Choice Parameter
- Credentials Parameter
- File Parameter
- Multi-line String Parameter
- Password Parameter
- Run Parameter
- String Parameter

Add Parameter ^

throttle builds-4 : هنا مثلا بحدده انها خلال ساعة او وقت معين يقدر يعمل run كام مره



☒ Throttle builds ?

Number of builds ?

1

Approximately 60 minutes between builds

Time period ?

Second

☐ Allow user triggered builds to skip the rate limit ?

concurrent-5 : هنا مثلا لو ضغطت علي pipeline مرتين ورا بعض يعملها run مرتين فنفس الوقت لكن لو مفعلتهاش فلو ضغطت مرتين هيعمل run للضغطه الاول ويستني تخلص ويبدأ في الثانية

Configure

General

Source Code Management

Source Code Management

None

Git ?

هنا لو عاوز ال pipeline يستدعي code من ال GitHub

Dashboard > automation > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Triggers

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

☐ Build periodically ?

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

بيحدد ال pipeline هيتعمله start ازاي وفيه اكثر من حاجة

trigger-1 : بتعمله run عن طريق script

after other project-2 : يتعمله run بعد pipeline معين وفيه اكثر من حاجة انه يشتغل لو كان ال pipeline الأول stable ولا unstable ولا failed او لو حصله aborted

☒ Build after other projects are built ?

Projects to watch

No project specified

☒ Trigger only if build is stable

☐ Trigger even if the build is unstable

☐ Trigger even if the build fails

☐ Always trigger, even if the build is aborted

periodically-3 : دي شبهه ال cronjob في ال Linux انه بتحددله وقت معين يتعمله run فيه بس هنا هيتعمل run لل pipeline كله

☒ Build periodically ?

Schedule ?

 No schedules so will never run

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

hook trigger-4 : هنا مثلا لو لما حد يعمل push علي ال repo ال GitHub بيعت زي notification فال pipeline دا يتعمله run والنوع دا مستخدم بكثرة

poll scm-5 : زي ال cronjob لكن هانا بيعمل check هل في تغييرات حصلت ولا لا في ال github لو حصلت اعمل run لل pipeline لو محصلش تغيير متعملش run

☒ Poll SCM ?

Schedule ?

No schedules so will only run due to SCM changes if triggered by a post-commit hook

☐ Ignore post-commit hooks ?

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Build Environment

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

في ال build environment في اكثر من حاجه

Delete workspace -1 : هنا كل مره بيعمل run لل pipeline بيحذف ال Folder ويعمله من اول وجديد

secret text-2 : دا لو علوز تحفظ credentials ك file

terminate-3 : يعني ال build لو حصله stuck يتعمله terminate ولا يفضل شغال

Build Steps

Execute shell ?

Command

See [the list of available environment variables](#)

```
ls
mkdir iti
cd iti
echo "hello Jenkins" > myfile
```

Advanced ▾

ال build steps دي الحاجات ال علوز انفذها (ال هي ال pipeline)

Post-build Actions

E-mail Notification

?

Recipients

Whitespace-separated list of recipient addresses. May reference build parameters like \$PARAM. E-mail will be sent when a build fails, becomes unstable or returns to stable.

mostafaamahmoud60@gmail.com

☒ Send e-mail for every unstable build

?

☐ Send separate e-mails to individuals who broke the build

Save

Apply

بعد ال build مثلا ممكن تخليه بيعت notification علي ال email

Dashboard > All >

Enter an item name

pipeline

» Required field

**Freestyle project**

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Pipeline**

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

**Multi-configuration project**

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

هعمل Create للنوع الثاني وهو ال Pipeline

Configure

-  General
-  Advanced Project Options
-  Pipeline

General

Enabled 

Description

Plain text [Preview](#)

- ☐ Discard old builds ?
- ☐ Do not allow concurrent builds
- ☐ Do not allow the pipeline to resume if the controller restarts
- ☐ GitHub project
- ☐ Pipeline speed/durability override ?
- ☐ Preserve stashes from completed builds ?
- ☐ This project is parameterized ?
- ☐ Throttle builds ?

Build Triggers

- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ GitHub Branches
- ☐ GitHub Pull Requests ?
- ☐ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?
- ☐ Quiet period ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

Advanced Project Options

Advanced 

Pipeline

Definition

Pipeline script 

Script ?

```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Hello') {
6       steps {
7         echo 'Hello world'
8       }
9     }
10  }
11 }
12
```

- ☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save

Apply

Configure

General

Advanced Project Options

Pipeline

☒ Discard old builds ?

Strategy

Log Rotation

Days to keep builds

if not empty, build records are only kept up to this number of days

Max # of builds to keep

if not empty, only up to this number of build records are kept

Advanced ▾

discard old builds-1 : بتحدد عدد او أيام معينه يسبلك فيها ال old build ودي حاجه مفيدة لأنه لو فضل يسبب كل ال old build ع المدي البعيد هيعمل مشكله في ال storage مثلاً هتتلمي بسرعه لأنها فالأخر هي files بتترمي ع ال Jenkins

وممكن اختار عدد أيام احتفظ بال build او احتفظ ب عدد build معين

Configure

General

☒ Do not allow concurrent builds☐ Abort previous builds ?

Do not allow concurrent builds : بمعنى ان مفيش build 2 يتعملهم run في نفس الوقت

General

☐ Do not allow the pipeline to resume if the controller restarts

: Do not allow the pipeline to resume if the controller restarts

هحددله مثلاً لو بيحصل build وحصل ريستارت لما ال Jenkins يرجع يكمل build من مكان ما وقف ولا يرجع يعيد من الأول

General

Advanced Project Options

Pipeline

☒ GitHub project

Project url ?

Advanced ▾

دا URL الخاص بال Project بتاعي علي ال Github

Configure

General

Advanced Project Options

☒ Pipeline speed/durability override ?

Custom Pipeline Speed/Durability Level

Maximum durability but slowest (previously the only option)

Performance-optimized: much faster (requires clean shutdown to save running pipelines)

Less durability, a bit faster (specialty use only)

Maximum durability but slowest (previously the only option)

: Pipeline speed/durability override

بحدله انا مهتم بالسرعة ولا الأداء الخاص بال Pipeline

☒ Preserve stashes from completed builds ?

Keep this many of the most recent builds' stashes ?

1

: Preserve stashes from completed builds

بقوله احفظلي عدد معين من ال Completed builds

: This project is parameterized

دا مثلا قبل م تعمل run لل pipeline يكون فيه option بتختاره

Filter

- Boolean Parameter
- Choice Parameter
- Credentials Parameter
- File Parameter
- Multi-line String Parameter
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Add Parameter ^

☒ Throttle builds ?

Number of builds ?

1

Approximately 60 minutes between builds

Time period ?

Second

☐ Allow user triggered builds to skip the rate limit ?

: Throttle builds

هنا مثلا بحدده انها خلال ساعة او وقت معين يقدر يعمل run كام مره

Configure

General

Advanced Project Options

Pipeline

Pipeline

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent any  
3  
4   stages {  
5     stage('Hello') {  
6       steps {  
7         echo 'Hello world'  
8       }  
9     }  
10  }  
11 }  
12
```

آخر حاجه بكتب ال Script الخاص بال Pipeline وبيكتب بلغة اسمها Groovy وهي لغة مخصصة
بال Jenkins

✓ Console Output

```
Started by user Admin  
[Pipeline] Start of Pipeline  
[Pipeline] node  
Running on Jenkins in /var/jenkins_home/workspace/pipeline  
[Pipeline] {  
[Pipeline] stage  
[Pipeline] { (Hello)  
[Pipeline] echo  
Hello World  
[Pipeline] }  
[Pipeline] // stage  
[Pipeline] }  
[Pipeline] // node  
[Pipeline] End of Pipeline  
Finished: SUCCESS
```

ودا ال Output الخاص بال Pipeline

Definition

Pipeline script from SCM

SCM ?

None

None

Git

Script Path ?

Jenkinsfile

☒ Lightweight checkout ?

[Pipeline Syntax](#)

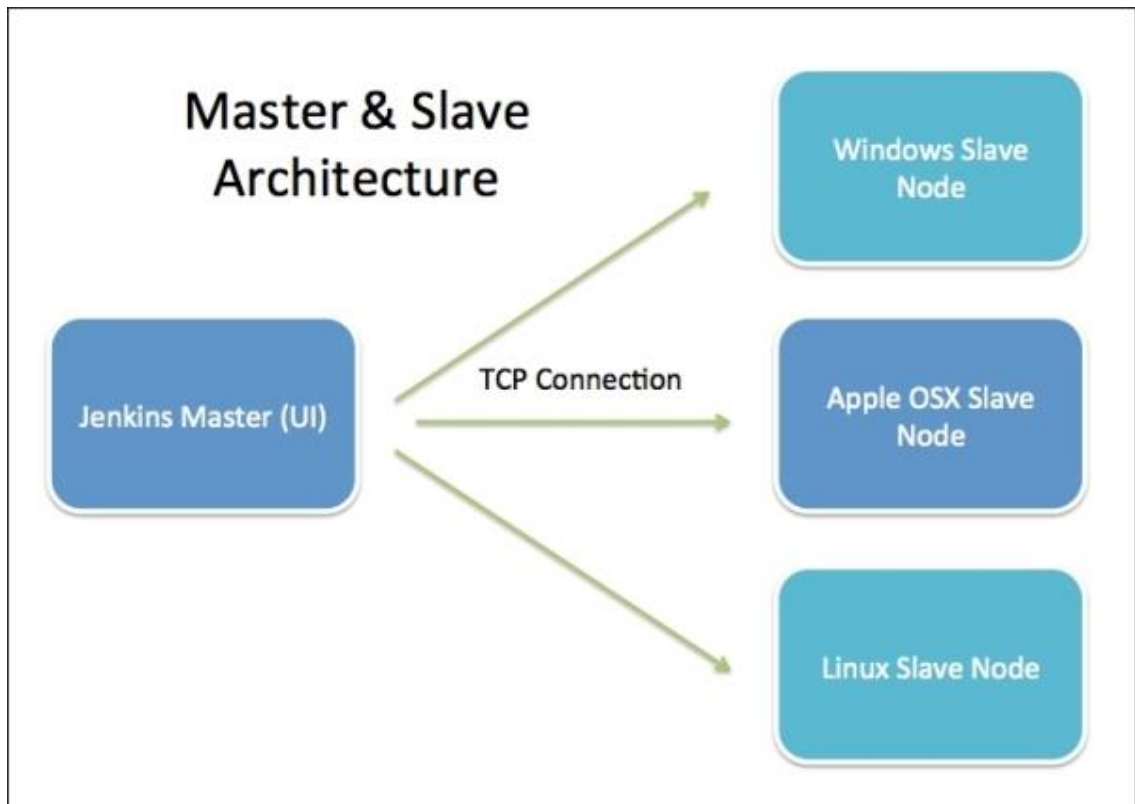
Save

Apply

دا النوع الثاني كتابه ال Pipeline : بقوله مش هعمل Script

انا هجيب Jenkinsfile جهاز مكتوب فيه ال Script بتاعي من ال Git

: Jenkins Architecture



بيكون عندي Jenkins Master ودي ال بعمل عليها ال Configuration الخاصة بال Jenkins

وبيكون عندي Agent ودا الي هعمل عليها ال Build

لازم يكون ال agent دا عليه :

1- ال JDK

2- Directory لل Workspace

3- Enable SSH

4- ال public Key الخاص بال Master

ال Jenkins عامله image علي ال Docker Hub مخصصه لل Agent وكل حاجه عليها جاهزة الا انك فقط هتضع ال Public Key الخاص بال Master تبعك

طيب ازاى اجيب ال Public Key بتاع ال Master ؟

```
[root@192 ~]# docker exec -it -u jenkins jenkins bash
jenkins@0a21919beccc:/$
```

1- هتدخل علي ال Container

--

```
jenkins@0a21919beccc:/$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/var/jenkins_home/.ssh/id_rsa):
Created directory '/var/jenkins_home/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /var/jenkins_home/.ssh/id_rsa
Your public key has been saved in /var/jenkins_home/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:vohgLkBGYbIGJc3mcQted0dMPHJ9neVH0zu5JGYfpNw jenkins@0a21919beccc
The key's randomart image is:
+---[RSA 3072]-----+
|+*o.. ..o. . . oo|
|++* o..=+ . . oo.|
|o= = . oo. . . +o|
|.oo . . . . + =|
|o      S      * E |
|.      .      o + +|
|. o      .      o |
|.o . . . . . |
|. . . . . |
+-----[SHA256]-----+
```

هتكتب ال Command الخاص بال Key ال هو ssh-keygen

```
[root@192 ~]# docker cp jenkins:/var/jenkins_home/.ssh/id_rsa.pub .
Successfully copied 2.56kB to /root/.
[root@192 ~]# ls
anaconda-ks.cfg  Downloads  ngrok-v3-stable-linux-amd64.tgz  Templates
Desktop          id_rsa.pub  Pictures                          Videos
Documents        Music      Public
```

بعد كذا انقل من ال Container للجهاز عادي بال docker cp

```
[root@192 ~]# docker run -d --name=agent --publish 2200:22 -e "JENKINS_AGENT_SSH_PUBKEY=$(cat id_rsa.pub)" jenkins/ssh-agent
Unable to find image 'jenkins/ssh-agent:latest' locally
latest: Pulling from jenkins/ssh-agent
fea1432adf09: Pull complete
a235bee2ae34: Pull complete
88736e9b2246: Pull complete
e3c311160846: Pull complete
4f4fb700ef54: Pull complete
6649848c81ad: Pull complete
bb9cc473595f: Pull complete
af03866356a6: Pull complete
Digest: sha256:2be1131e2f5df0d595b23550849fc4f57f6261f76166618c98c45ad573ab9178
Status: Downloaded newer image for jenkins/ssh-agent:latest
2290e5dace55e525b74d1e314266f6c4a149ac2358b2e87f44154c9b4a09cdde
```

واخير هتعمل Run لل image الخاصه بال agent

Jenkins

Dashboard > Manage Jenkins > Nodes >

Nodes

+ New Node

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	63.02 GiB	3.92 GiB	63.02 GiB	0ms
	Data obtained	52 min	52 min	52 min	52 min	52 min	52 min

Icon: S M L

Legend

هنتفتح ال Master وتدخل علي Manage Jenkins ثم Node ثم New Node

Dashboard > Manage Jenkins > Nodes > New node

New node

Node name

Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

هتكتب اسم لل Agent وضغط Create

agent-01

Description ?

Plain text [Preview](#)

Number of executors ?

1

ال Number of executors هي هسمح لكam Build يحصل في نفس الوقت

Remote root directory ?

! Remote directory is mandatory

ال Remote root directory : Dir الخاص بال Workspace
طيب اجيبه منين؟ ممكن تعمل docker Inspect لل agent هتلاقيه موجود

```
"/var/run": {}  
},  
"WorkingDir": "/home/jenkins",  
"Entrypoint": [  
    "setup-sshd"  
],  
"ExposedPorts": {}  
}
```

Remote root directory ?

/home/jenkins

Labels ?

agent-01

Usage ?

Only build jobs with label expressions matching this node

ال labels بميز ال agent دا

ال usage بقوله متعملش Run علي ال Agent دا غير لما ال Labels يساوي ال Label بتاع ال Agent دا

Launch method ?

Launch agents via SSH

Host ?

172.17.0.3

Credentials ?

jenkins

+ Add ▼

ال Launch Method دي الطريقة ال هعمل بيها Connect علي ال Agent وهي ال SSH
ال Host ودا ال IP الخاص بال Container بتاع ال Agent وممكن تعرفه عن طريق ال inspect
ال Credentials وهنا بتضع ال Privet key الخاص بال Master

Host Key Verification Strategy ?

Non verifying Verification Strategy

Advanced ▾

Availability ?

Keep this agent online as much as possible

ال Verification لو عاوز تعمل Verification لما تعمل Connect
ال Availability هي ان ال Agent هيفضل متاح عطول ولا في وقت معين فقط

Dashboard
Nodes

Nodes

Clocks

Build Queue

No builds in the queue.

Build Executor Status

built-in node + 1 agent (0 of 2 executors busy)

Nodes

+ New Node

Configure Monitors

Refresh

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	agent-01	Linux (amd64)	In sync	62.65 GiB	3.92 GiB	62.65 GiB	774ms
	Build-In Node	Linux (amd64)	In sync	62.65 GiB	3.92 GiB	62.65 GiB	0ms
Data obtained		1 min 39 sec	1 min 39 sec	1 min 39 sec	1 min 39 sec	1 min 39 sec	1 min 39 sec

Icon:

S

M

L

Legend

بعد متعمل Save هتلاقي ال Agent ظهر معاك

```
INFO: Both error and output logs will be printed to /home
<==[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3206.vb_15dcf73f6a_9
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

لو دخلت عليه وفتحت ال Log هتلاقي انه عمل Connect لو الدنيا تمام

<u>Build Executor Status</u>		▼
 <u>Built-In Node</u>		
1	Idle	
2	Idle	
 <u>agent-01</u>		
1	Idle	

هتلاقيه ظهر في ال Build

<u>Build Executor Status</u>		▼
 <u>agent-01</u>		
1	Idle	

دخلت علي ال Master وخليت ال Number of executors بتساوي صفر

فال Build Executor Status مفهانش غير ال Agent مفيش Run هيجصل علي ال Master

Labels ?

agent-02

Usage ?

Use this node as much as possible

ضفت Agent كمان بس ال Usage سبتها عادي انه يحصل عليه Build مش لازم احدد ال Label بتاعه

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent any  
3  
4   stages {  
5     stage('Hello') {  
6       steps {  
7         echo 'Hello World'  
8       }  
9     }  
10  }  
11 }  
12
```

وبالتالي في ال Script لو سبت ال Agent يكون Any فيتهتمله Build علي ال Agent-02



Console Output

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent-02 in /home/jenkins/workspace/pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent {label 'agent-01'}  
3  
4   stages {  
5     stage('Hello') {  
6       steps {  
7         echo 'Hello World'  
8       }  
9     }  
10  }  
11 }  
12
```

عشان اخليه يتعمله Run علي ال Agent-01 بحدده ان ال Labels بتاعه اسمه agent-01



Console Output

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent-01 in /home/jenkins/workspace/pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

لما اجي اعمل build هيحصل علي ال Agent-01

طيب لحد كذا تمام ؟ لو جينا استخدمنا في ال Script أوامر خاصه بال Docker مش هيشغل لان ال Agent معلوش ال Docker وعشان اشغل عليه ال Docker :

```
[root@192 ~]# mkdir jenkins-agent-with-docker
```

هعمل Dir جديد

```
[root@192 ~]# cd jenkins-agent-with-docker
```

هدخل في ال Dir

```
[root@192 jenkins-agent-with-docker]# vim Dockerfile
```

هعمل Dockerfile

```
FROM jenkins/ssh-agent
RUN apt-get update -y
RUN apt-get install -y ca-certificates curl gnupg lsb-release
RUN curl -fsSLo /usr/share/keyrings/docker-archive-keyring.asc \
  https://download.docker.com/linux/debian/gpg
RUN echo "deb [arch=$(dpkg --print-architecture) \
  signed-by=/usr/share/keyrings/docker-archive-keyring.asc] \
  https://download.docker.com/linux/debian \
  $(lsb_release -cs) stable" > /etc/apt/sources.list.d/docker.list
RUN apt-get update && apt-get install -y docker-ce-cl
```

هكتب ال Code بتاعي وال هو عبارته ال Agent عادي لكن بعمل install لل docker كمان

```
[root@192 ~]# docker run -d --name=agent3 --publish 2400:22 -e "JENKINS_AGENT_SSH_PUBKEY=$(cat id_rsa.pub)" -v /var/run/docker.sock:/var/run/docker.sock agent3
ssh_docker
0285d1dd481b9cd345c9b337832adbf041e5618dd427541a76d9a62a37677da
```

بعد كذا واخر حاجه بعمل ال Comannd بتاعي عادي

لكن ملحوظة لازم تعمل Volume ال هو /var/run/docker.sock:/var/run/docker.sock
عشان تعرف تستخدم ال Commands الخاصه بال Docker بال Script الخاص بال Jenkins

Script ?

```
1 pipeline {
2   agent { label 'agent-docker' }
3   environment {
4     PATH = "/usr/bin:/usr/local/bin:$PATH"
5   }
6   stages {
7     stage('Test Docker') {
8       steps {
9         sh 'docker --version'
10      }
11    }
12  }
13 }
14 }
15 }
```

في ال script بحدد ال env بتاعتي

وعاوز انفذ ال Command دا `docker --version`



Console Output

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent-03 in /home/jenkins/workspace/dockerpipeline
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Test Docker)
[Pipeline] sh
+ docker --version
Docker version 27.0.3, build 7d4bcd8
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

فهتلاقى انه تمام وعمل ال Command بتاعتي

Slack Notification دي Plugins بتبعناك Message مثلا بعد ال Run لو ال Build اتعمل اتبعناك رسالة معينه وهكذا

طيب اعمله Config ازاي ؟

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Slack Notification

Loading plugin extensions

Pending

Pending

→ [Go back to the top page](#)

(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

هعمل Install لل Plugins واسمها Slack Notification

هعمل Account علي ال Slack

وبعد كذا هعمل Workspace

وبعد كذا هعمل Channel



Search

Documentation Tutor

Your Apps

Build something amazing.

Use our APIs to build an app that makes people's working lives better. You can create an app that's just for your workspace or create a public Slack App to list in the App Directory, where anyone on Slack can discover it.

Create an App

بعد كذا هروح علي api.slack.com/apps

وهعمل Create ل App

Name app & choose workspace



App Name

Jenkins

Don't worry - you'll be able to change this later.

Pick a workspace to develop your app in:



DevOps-in-Arabic



Keep in mind that you can't change this app's workspace later. If you leave the workspace, you won't be able to manage any apps you've built for it. The workspace will control the app even if you leave the workspace.

[Sign into a different workspace](#)

By creating a **Web API Application**, you agree to the [Slack API Terms of Service](#).

Cancel

Create App

بعمل اسم لل App وبختار ال Workspace ال عملتها create

Incoming Webhooks

Post messages from external sources into Slack.

Interactive Components

Add components like buttons and select menus to your app's interface, and create an interactive experience for users.

Slash Commands

Allow users to perform app actions by typing commands in Slack.

Event Subscriptions

Make it easy for your app to respond to activity in Slack.

Bots

Allow users to interact with your app through channels and conversations.



Permissions

Configure permissions to allow your app to interact with the Slack API.

هيدا ادي ال App دا Permissions

Scopes

A Slack app's capabilities and permissions are governed by the [scopes](#) it requests.

Bot Token Scopes

Scopes that govern what your app can access.

OAuth Scope	Description ↓	
chat:write	Send messages as Jenkins	
users:read.email	View email addresses of people in a workspace	
users:read	View people in a workspace	

Add an OAuth Scope

ال Permission بتكون في ال Scops وظيف ال محتاجه من ال Permissions

Features

App Home
Org Level Apps
Incoming Webhooks
Interactivity & Shortcuts
Slash Commands
Workflow Steps
OAuth & Permissions
Event Subscriptions
User ID Translation
App Manifest
Beta Features

Submit to App

opted into token rotation

Opt In

OAuth Tokens for Your Workspace

These OAuth Tokens will be automatically generated when you finish connecting the app to your workspace. You'll use these tokens to authenticate your app.

Install to Workspace

بعد کدا بتعمل Install لل App دا ال Workspace



Jenkins is requesting permission to access the DevOps-in-Arabic Slack workspace

What will Jenkins be able to view?



Content and info about your workspace



What will Jenkins be able to do?



Perform actions in channels & conversations



Cancel

Allow

تعمل Allow

OAuth Tokens for Your Workspace

These tokens were automatically generated when you installed the : You can use these to authenticate your app. [Learn more.](#)

Bot User OAuth Token

xoxb-7384110077810-7384177063298-evFo0Anf9TkKFYRpcaF8

Access Level: Workspace

Reinstall to Workspace

هيطلعلك ال Tokens دا ودا ال هتخده وتضعه في ال Jenkins

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >

Secret text

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Secret

.....

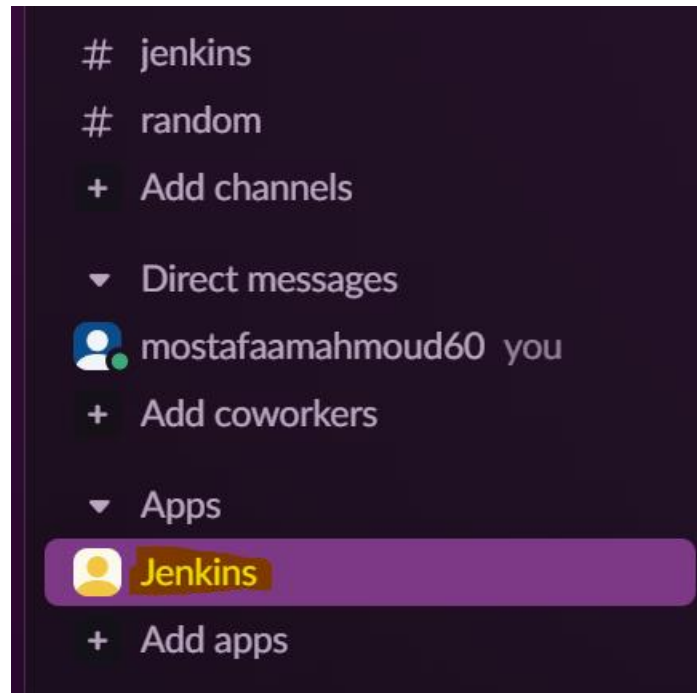
ID ?

slack

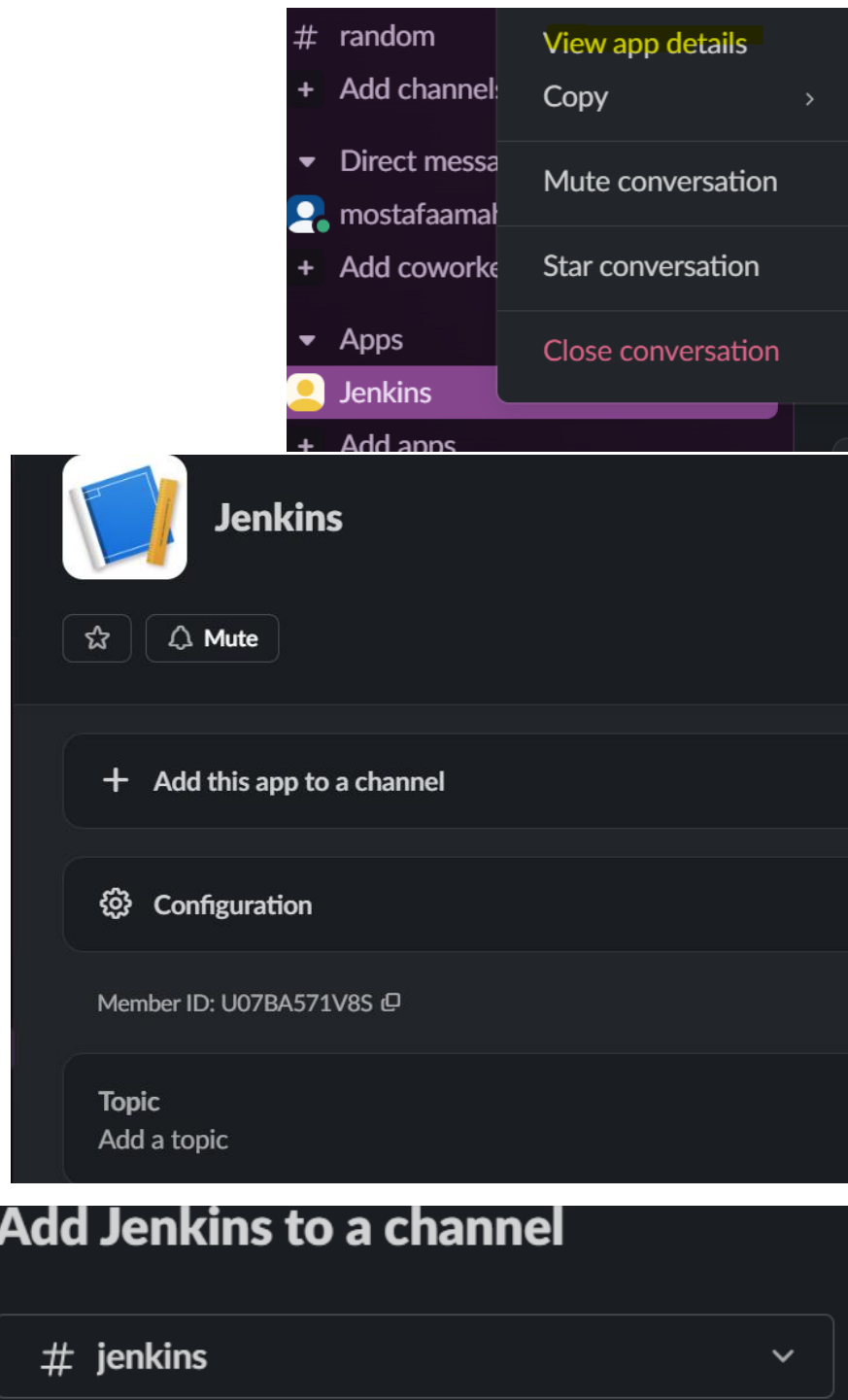
Description ?

Create

هتروح علي ال Jenkins وتضيف Credentials وبتكون من نوع Secret



تتأكد ان ال App ال عملته مضاف داخل ال Channel



ف العاده مش بيكون مضاف ف انت بتضيفه وتتأكد

Slack

Workspace ?

DevOps-in-Arabic

Credential ?

slack

+ Add ▾

Default channel / member id ?

#jenkins

☒ Custom slack app bot user ?

Advanced ▾

Save

Apply

آخر خطوه هي ال Config بتاع ال Slack علي ال Jenkins
من Manage Jenkins هتدخل علي System هتلاقي Section خاص بال Slack تبدا تضيف ال Config دي

ال Workspace هي اسم ال Workspace علي ال Slack
ال Credential هي ال Tokens ال طلعتنا لما عملنا ال App علي ال Slack
ال Channel هي اسم ال Channel علي ال Slack ولازم يكون قبلها ال #

+ Add ▾

Default channel / member id ?

#jenkins

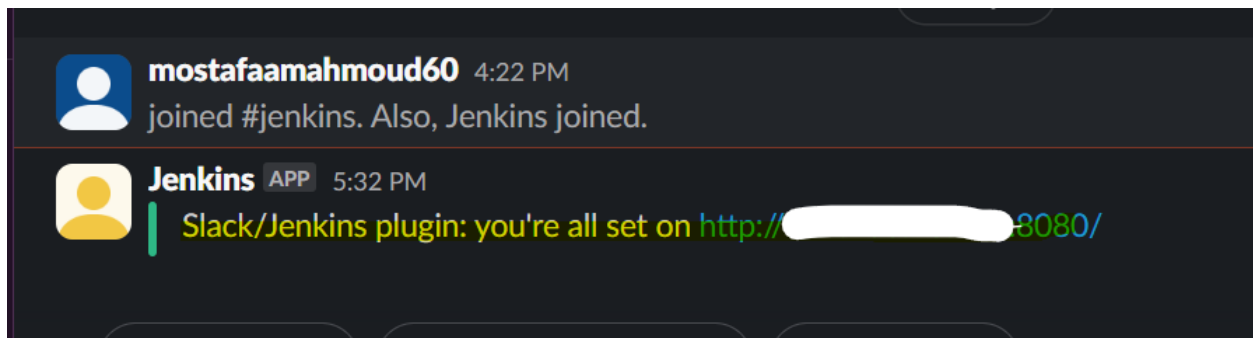
☒ Custom slack app bot user ?

Advanced ▾

Success

Test Connection

لو عملت Test connection وال Config مضبوطه هيجبك Success



بعد م تعمل Save لو روجت علي ال Slack هتلاقي في message انك جاهز تستقبل

طيب كدا دي ال Config ازاي ابدا استخدمها ؟

Script ?

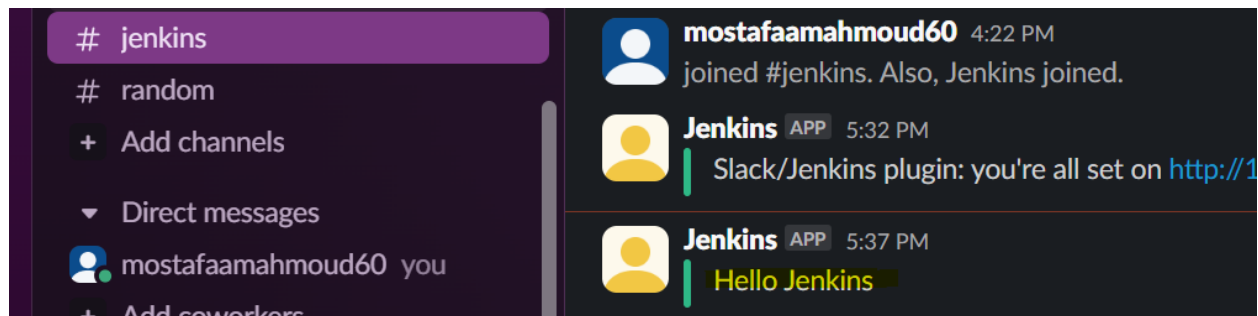
```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Hello') {
6       steps {
7         echo 'Hello World'
8       }
9       post {
10        success {
11          slackSend color: "good", message: "Hello Jenkins"
12        }
13      }
14    }
15  }
16 }
17
```

بالطريقه دي فال Script بقوله لو ال Build كان Success ابعثلي message علي ال Slack تقولي فيها
Hello Jenkins

Console Output

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent-02 in /home/jenkins/workspace/pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
Post stage
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: DevOps-in-Arabi
color: good, botUser: true, tokenCredentialId: slack, notifyCommitters: false, iconEmoji: <e
<empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

لما تعمل Run ويكون Success في ال Logs هيقولك انه بعثلك ال message علي ال Slack



ولو روجت علي ال Slack هتلاقي ال message ال كتبناها في ال Pipeline Script

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